Phonocardiogram (PCG) Classification Using a 1D Convolutional Spiking Neural Network (CSNN)

A report submitted in partial fulfillment of the requirements for
the award of the degree of

Bachelor of Technology

in

Department of Electronics and Communication Engineering

by

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May 2026

BONAFIDE CERTIFICATE

This is to certify that the project work entitled "**Phonocardiogram (PCG) Classification Using a 1D Convolutional Spiking Neural Network (CSNN)**" is a Bonafide record of the work done by

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CHAPTER 1

ABSTRACT

Despite advancements in cardiac care, early detection of Valvular Heart Diseases (VHD) remains a challenge due to the "power-accuracy trade-off" in wearable technology. Traditional deep learning models for Phonocardiogram (PCG) analysis are often too computationally intensive for continuous monitoring, leading to rapid battery depletion in edge devices. This project addresses the critical need for a high-accuracy, ultra-low-power diagnostic framework that can operate autonomously on wearable hardware, ensuring that life-threatening cardiac irregularities are identified long before they reach a critical stage.

Our approach shifts from conventional artificial neural networks to Spiking Neural Networks (SNNs), which mimic the brain's event-driven processing to achieve extreme energy efficiency. We developed a comprehensive pipeline that integrates advanced signal pre-processing with a ConvSNN1D architecture. By systematically evaluating five distinct feature extraction "encoders," we identified the Constant-Q Transform (CQT) as the superior representation for spiking neurons, achieving a peak validation accuracy and a balanced sensitivity and specificity.

The significance of this work lies in its ability to maintain high-fidelity diagnostic performance on noisy, real-world datasets using a remarkably lean architecture of very low parameters. By overcoming the limitations of previous models that suffered from performance degradation in environmental noise, our framework proves that sophisticated medical screening can be both non-invasive and computationally efficient.

ACKNOWLEDGEMENTS

We would like to express a deep sense of gratitude to Lord Almighty for giving us an opportunity to do this project and showering his blessing in the due course of the project.

We are grateful to our project supervisor, **Dr. Suresh Balanethiram** for their unwavering support, insightful feedback, and expert guidance throughout the project. Their expertise and encouragement have been instrumental in the successful completion of this work.

We are grateful to **Dr. Makarand Madhao Ghangrekar**, Director, NIT Puducherry, for providing a conducive environment for learning and research. I extend my heartfelt thanks to **Dr. Sundaravarathan,** Registrar, NIT Puducherry, for the administrative support throughout the project.

I would like to express my gratitude to the Dean, Academics of NIT Puducherry, for the valuable insights, encouragement, and support during this project. A special word of thanks goes to **Dr.Aniruddha Kanhe**, The Head of the Department of Electronics and Communication for her continuous guidance, mentorship, and encouragement. Thanks are due to our project coordinator**, Dr. Malaya Kumar**. His expertise and knowledge have been invaluable in shaping the direction of this project.

We would also like to acknowledge the faculty members, mentors, and colleagues who provided valuable suggestions, feedback, and assistance during the course of this project. Their contributions have greatly enriched the project.

CHAPTER 2

LIST OF ABBREVATION

1. **ANN**: Artificial Neural Networks
2. **AUC-ROC**: Area Under the Receiver Operating Characteristic
3. **CAD**: Coronary Artery Disease
4. **CNN**: Convolutional Neural Networks
5. **CQT**: Constant-Q Transform
6. **CSNN**: Convolutional Spiking Neural Network
7. **CVD**: Cardiovascular Disease
8. **CWT**: Continuous Wavelet Transform
9. **dB**: Decibels
10. **DC**: Direct Current
11. **DCT**: Discrete Cosine Transform
12. **DSP**: Digital Signal Processing
13. **ECG**: Electrocardiogram
14. **FPGA**: Field Programmable Gate Array
15. **HIL**: Hardware-in-the-Loop
16. **IIR**: Infinite Impulse Response
17. **LIF**: Leaky Integrate-and-Fire
18. **MFCC**: Mel Frequency Cepstral Coefficients
19. **MVP**: Mitral Valve Prolapse
20. **PCG**: Phonocardiogram
21. **PZT**: Lead Zirconate Titanate (piezoelectric sensor)
22. **RTL**: Register Transfer Level
23. **S1, S2**: First and Second Heart Sounds
24. **SNN**: Spiking Neural Network

25. **SNR**: Signal-to-Noise Ratio
26. **STFT**: Short-Time Fourier Transform
27. **VHD**: Valvular Heart Disease
28. **XAI**: Explainable Artificial Intelligence

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CHAPTER 3
INTRODUCTION

The Phonocardiogram (PCG) is a digital recording of the acoustic energy produced by the mechanical activity of the heart, specifically capturing the vibrations caused by blood flow and the closure of the four cardiac valves. The primary advantage of PCG analysis lies in its ability to provide direct insight into the heart's structural integrity, as valvular pathologies like stenosis and regurgitation manifest as audible murmurs long before they impact the electrical rhythm. Furthermore, PCG signals are rich in non-stationary frequency components, making them exceptionally well-suited for high resolution feature extraction techniques.

The signal can be acquired non-invasively using a single acoustic sensor, eliminating the need for the multi lead configurations or conductive gels required. This combination of high information density and low hardware complexity makes the PCG an ideal candidate for cost-effective, portable health monitoring device with real-time screening and early detection of mechanical heart disease.

The integration of Artificial Intelligence and Machine Learning with Phonocardiogram (PCG) signal analysis transforms traditional diagnostic methods into an intelligent, adaptive system capable of identifying high dimensional, non-linear patterns that remain invisible to manual inspection. The primary benefit of this approach is automated feature extraction, allowing deep learning models to independently discover critical diagnostic markers within complex acoustic data, eliminating the limitations of human-defined rules. AI driven systems demonstrate exceptional robustness against real-world environmental noise ensuring high diagnostic accuracy in non-clinical settings. By leveraging historical data to establish sophisticated baselines, these models offer superior scalability and precision, enabling early-stage screening of valvular diseases at an expert level while maintaining the computational efficiency required for real-time monitoring.

Spiking Neural Networks (SNNs) are utilized for signal analysis primarily due to their biologically inspired architecture, which processes information through discrete, timed events known as spikes rather than continuous numerical values. The fundamental advantage of this model is its extreme energy efficiency, as the event driven nature ensures that neurons remain idle and consume minimal power in the absence of input, making it ideal for battery-operated hardware. By encoding data in the precise timing of pulses, SNNs inherently capture the temporal dynamics and rhythmic patterns of time series signals with high fidelity. Furthermore, the use of binary spikes replaces floating-point multiplications with simple additions, significantly reducing computational overhead and memory requirements. This combination allows SNNs to provide high-performance, real-time processing with power constraints of edge computing and neuromorphic systems.

CHAPTER 4
STATE-OF-THE-ART

This paper **[1]** focused on spiking neural network and its efficiency. By utilizing **Spiking Neural Networks (SNNs)**, which mimic the brain's event driven processing, the authors achieved high accuracy in cardiovascular disease (CVD) diagnosis while significantly reducing power consumption. The model provided **High Energy Efficiency** first of its kind for ECG+PCG, using 11.9x less energy than ANNs and reduced noise without extra computational overhead. But it was **ECG over reliant** and provided binary classification alone hence limiting the models edge device implementation heavily.

While to overcome this limitation another paper **[12]** proposed a dual-domain approach. Instead of relying solely on time series data, it transforms **Phonocardiogram (PCG)** signals into the time frequency domain using methods like **STFT (Short Time Fourier Transform)** or **Wavelets**. This allowed CNN based framework to visualise the spectral components of heart murmurs that were invisible in the time domain. This method provided around ~100% for 5 class heart valve disease classification. Polynomial Chirplet transform captured complex PCG features better than traditional methods. Since PCT+CNN pipeline was too complex for low power wearable further its accuracy dropped by ~85% on noisy datasets like PhysioNet making it incompatible for edge application.

Even though it was challenging to implement an edge device friendly model, a new method was being introduced which focused on Explainable AI (XAI) and Clinical Relevance **[5]**. Rather than just predicting a disease, this study focused on **Explainable AI**. By identifying specific event patterns associated with Mitral Valve Prolapse (MVP). The framework was designed to provide clinically relevant feedback, explaining **why** a specific diagnosis was made. This model was highly successful due to its ability to provide clinically interpretable reasoning by precisely focusing on clicks/murmurs and isolating fine-grained heart events (S1, S2) without needing ECG. But it fell short since it was validated on a very small, private dataset (124 subjects) and it wasn't edge optimized as well.

This particular paper **[8]** shifts from offline analysis to more active monitoring. It details the development of a **wearable sensor** that captures both electrical (ECG) and acoustic (PCG) cardiac data simultaneously. The integration allows for a more holistic view of cardiac health, such as calculating the **Electromechanical Systole**. At first, it's quite interesting to note the salient feature of achieving 3dB SNR signal quality with custom wearable PZT sensor along with full end-to-end system from data acquisition to mobile app. But the processing had to be done offline due to absence of automated classifier, and its focus was on CAD rather than VHD.

4.1 Dataset

The PhysioNet/CinC 2016 Challenge dataset serves was the primary data source used the project, comprising a total of 3,153 heart sound recordings sourced from 764 distinct subjects. The data is structured into six training folders (Folders A to F), each representing a different clinical origin and research group. These folders vary significantly in size and composition: Folder E is the largest with approximately 2,141 recordings, followed by Folder A (409), Folder B (490), Folder C (31), Folder D (55), Folder F (114). Each folder contains unique characteristics such as diverse patient populations, varying signal durations from 5 to 120 seconds, and different recording hardware ranging from traditional acoustic to digital electronic stethoscopes.

The recordings within these folders differ primarily in their noise profiles and pathological signatures along with heavy ambient interference like speech or breathing sounds, providing a benchmark for real-world application.

4.2 Objective of Thesis

1. To design and implement a **1D Convolutional Spiking Neural Network (CSNN)** architecture.
2. Classifying Phonocardiogram (PCG) signals into normal and abnormal categories.
3. Model should provide high computational and energy efficiency suitable for edge-device integration.

CHAPTER 5
METHODOLOGY

5.1 PROPOSED PIPELINE

Raw PCG Signal → **Pre-processing** → **Feature Extraction** → **Spike Encoding** → **1D CSNN** → **Evaluation Metrics**

FLOW CHART – 1 PIPELINE

The processing pipeline (**FLOW CHART 1**) transforms raw acoustic heart sounds into actionable diagnostic insights through a structured five-stage sequence:

1. **Pre-processing**: Raw PCG signals undergo DC removal, notch filtering, and bandpass filtering (25–400 Hz) to eliminate baseline wander and powerline interference.
2. **Feature Extraction**: The denoised signal is analyzed using specialized time-frequency representations, such as the **Constant-Q Transform (CQT)** or **Mel-spectrograms**, to highlight spectral patterns invisible in the time domain.
3. **Spike Encoding**: Extracted features are converted into discrete temporal events (spikes) using **Leaky Integrate-and-Fire (LIF)** neurons, mimicking biological neural processing.
4. **1D CSNN**: A lean, **120,000-parameter Convolutional Spiking Neural Network** processes these spike trains to identify complex diagnostic patterns with high energy efficiency.
5. **Evaluation Metrics**: The final output is assessed using **Validation Accuracy**, **Sensitivity**, and **Specificity** to ensure reliable multi-class classification for heart valve disorders.

5.2 RAW PCG SIGNAL:

PCG signals generally are recorded at 8 kHz to protect signal integrity at hardware level i.e. protection against aliasing during analog to digital conversion, capturing unexpected high freq component.

The signal is down sampled to 4kHz as Nyquist freq of 2 kHz of down sampled signal is enough to represent all the relevant PCG component.

Down sampling also helps in reducing the computational load and simplifies filter design.

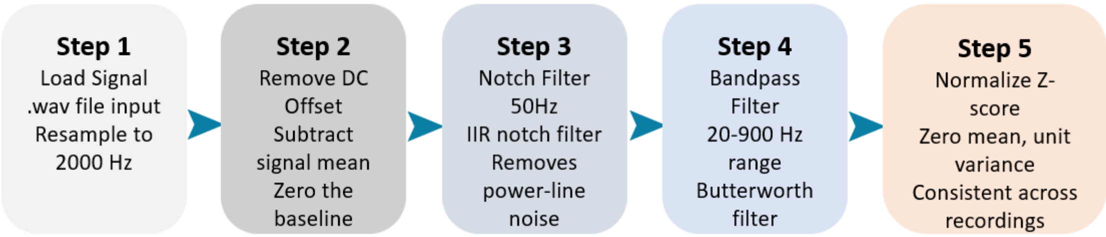
Clinically Significant information of PCG signal lies in the range of 5-1000 Hz with:

Range of S1: 10-50 Hz.

Range of S2: 20-250 Hz.

Range of murmur: 20-600 Hz.

5.3 SIGNAL PRE-PROCESSING:



FLOW CHART – 2 PRE-PROCESSING

The down-sampled signal is pre-processed in this step for transformation required for feature extraction.

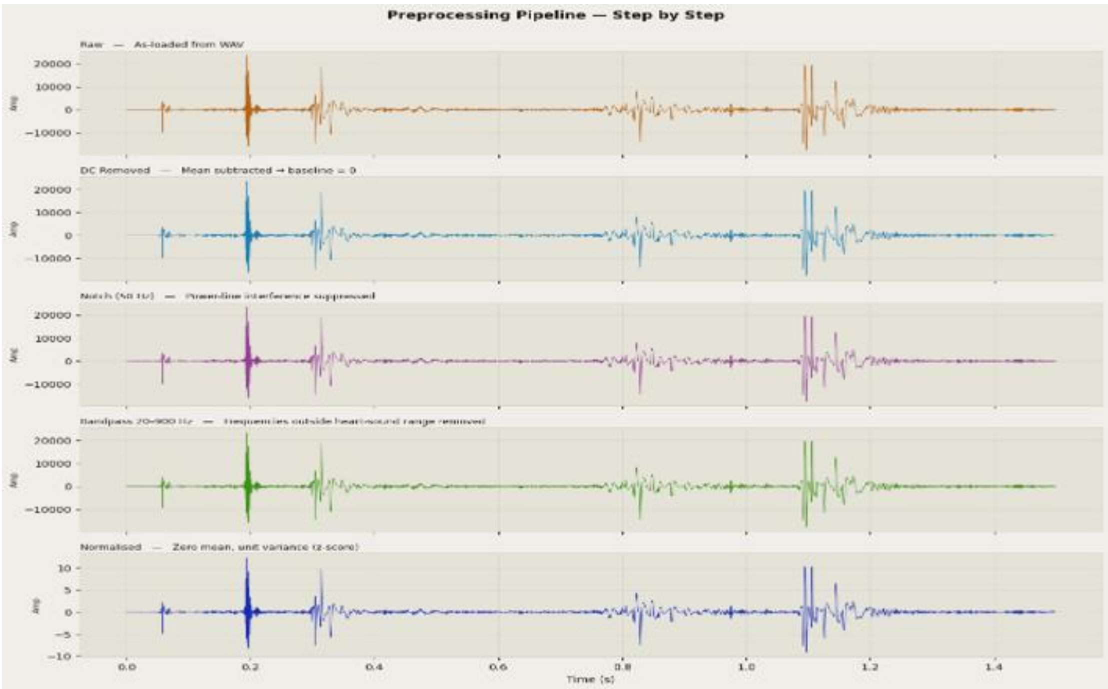


Fig.1 PRE PROCESSING

First as shown in FLOW CHART 2 , we remove the DC offset i.e calculate the mean of the signal and subtract from every sample to effectively zero the baseline. This is done since raw recordings often have non-zero avg due to electronic equipment bias. This step ensures the signal oscillates around a true zero point.

Then we pass this signal thru a IIR Notch filter with cutoff at 50Hz. This is done to remove “power-line noise” or hum caused by electrical interference from the AC power grid, which typically occurs at 50 Hz (or 60 Hz in some regions).

Once after passing the signal thru notch filter to remove murmur signal is passed thru a low pass filter (bandpass filter) of 300 Hz as cutoff freq. This effectively removes baseline wanders below 20 Hz (like breathing or muscle movement) and above 900 Hz (high-frequency ambient noise), hence the model can focus only on the relevant diagnostic information.

Finally, the band passed signal is z-score normalised i.e signal is adjusted to have zero mean and unit variance. This is done cause PCG recordings vary greatly in volume depending on the microphone placement or the patient’s chest wall thickness. Normalization ensures that the neural network treats all recordings with the same scale, preventing loud recordings from having an undue influence over quiet ones.
The final output result would look as shown in **figure 1**

5.4 Feature extraction:

The PCG signal, shown in **figure 2**, on which 5 feature extraction methods will be executed namely, Mel Frequency cepstral co-efficient (**MFCC**), Mel spectrogram, Continues Wavelet transform (**CWT**), **Chirplet transform**, Constant Q transform (**CQT**), to find which one is optimal and best one that aligns with our project objective.
The details about the feature extraction methods and their features along with output obtained after transformation have been provided below.

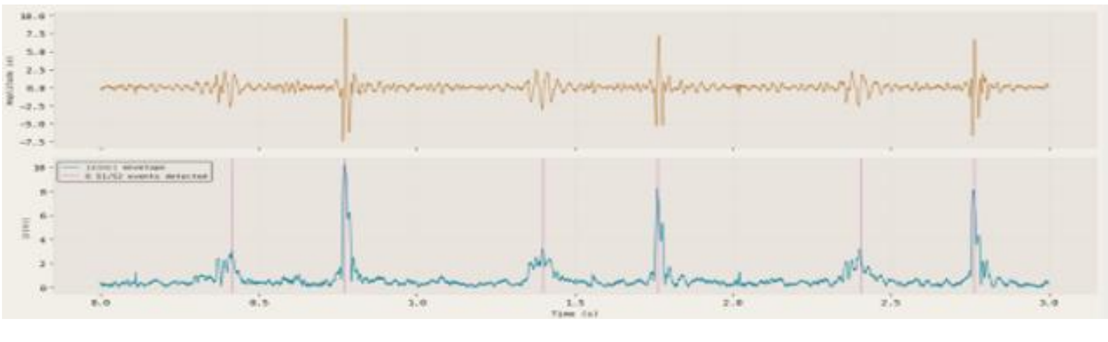


Fig.2 PRE-TRANSFORMED PCG SIGNAL

5.4.1 Mel Frequency Cepstral Co-efficient (MFCC):

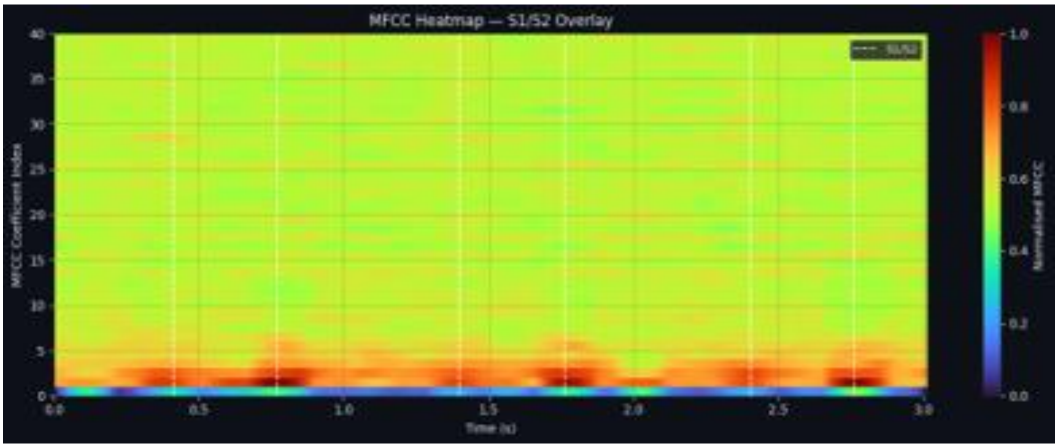


Fig.3 POST MFCC TRANSFORMED PCG

Widely used feature extraction method in cardiac signal processing.

They represent the "texture" of heart sounds in a way that mimics human auditory perception.

MFCCs use a non-linear **Mel scale**, which provides higher resolution for the lower frequencies where critical heart sound components like S1 and S2 reside.

The pre-processed signal is framed, windowed (usually 20-40 ms), then fft is applied, mel filter bank created, finally Discrete cosine transform (DCT) is applied to obtain MFCC (**figure 3**).

Formula used for this process are as follows:

1. $y(t) = x(t) - \alpha x(t-1)$→ pre-processed signal α ~ 0.97.
2. $w(n) = 0.54 - 0.46\cos\left(\frac{2\pi n}{N-1}\right)$→ hamming window function.
3. $P(k) = \frac{1}{N}\left|\sum_{n=0}^{N-1} x(n) e^{-j2\pi kn/N}\right|^2$→ Fast Fourier transform.
4. $M(f) = 2595 \cdot \log_{10}(1 + f/700)$→ Mel frequency.
5. $S(m) = \sum_{k=0}^{N-1} P(k) H_m(k)$→ Mel spectrum coefficient.
6. $C_q = \sum_{m=1}^{M} \log\left(S(m)\right) \cos\left[\frac{q(m-0.5)\pi}{M}\right]\Big)$→ DCT

Features:

Validation Accuracy: 73.6%

Sensitivity: 73.9%

Specificity: 63.9%

5.4.2 MEL SPECTROGRAM:

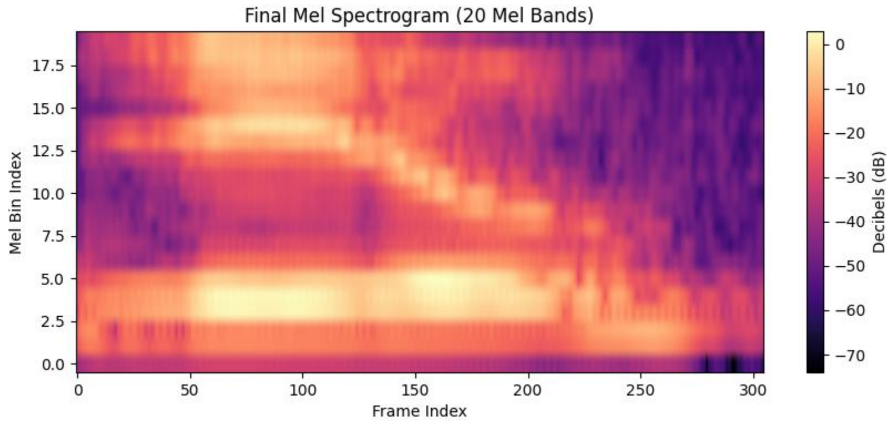


Fig.4 POST MEL TRANSFORMED PCG

Visual and mathematical representation of a signal's power spectrum over time, mapped onto the **Mel scale.**

It warps the frequency axis to match how the human ear perceives sound, hence prioritizing lower frequencies.

Processing steps are almost like MFCC **(figure 3)** except here STFT is used instead of FFT.

Equations used are as follows:

1. $w[n] = 0.54 - 0.46\cos\left(\frac{2\pi n}{N-1}\right)$→ Hamming window is applied to rescue signal discontinuities.
2. $X(m,k) = \sum_{n=0}^{N-1} x\,[n + mH]w[n]e^{-j2\pi kn/N}$→ STFT calculation.
3. $P(m,k) = |X(m,k)|^2$→ power spectrum calculation.
4. $M(f) = 2595 \cdot \log_{10}(1 + f/700)$→ Mel frequency conversion.
5. $S(m,p) = \sum_k P\,(m,k)H_p(k)$→ Mel filter bank integration.

Features:

Validation Accuracy: 93%

Sensitivity: 92%

Specificity: 92.5%

5.4.3 Continuous Wavelet Transform (CWT):

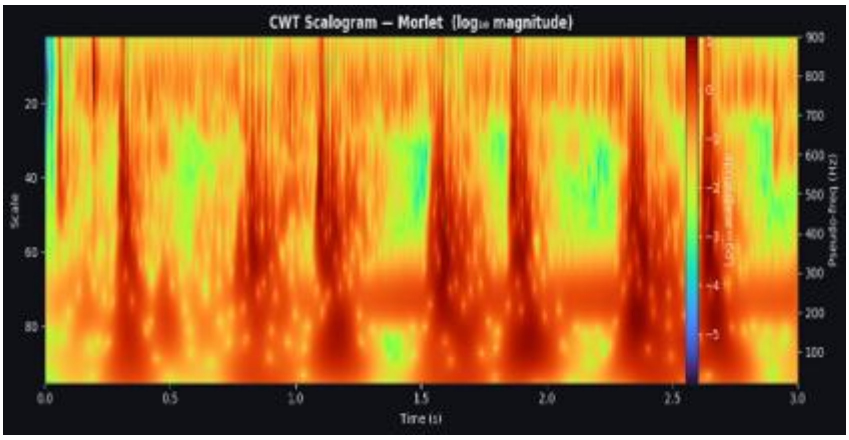


Fig.5 POST CWT TRANSFORMED PCG

Time-frequency analysis tool that is particularly effective for non-stationary signals like **PCG**. Uses a **"mother wavelet"**—a small, localized wave that is scaled (stretched) and translated (shifted) across the signal shown in **fig 5**.

Provides excellent time resolution for high frequencies and excellent frequency resolution for low frequencies.

Equations used are as follows:

1. $\Psi_{a,b}(t) = \frac{1}{\sqrt{a}}\Psi\left(\frac{t-b}{a}\right)$→ modified mother wavelet
2. $CWT_x(a,b) = \int_{-\infty}^{\infty} x\,(t)\frac{1}{\sqrt{a}}\Psi^*\left(\frac{t-b}{a}\right)dt$→ CWT equation.

Features:

Validation accuracy: 87.3%

Sensitivity: 88.6%

Specificity: 82.5%

5.4.4 Chirplet Transform:

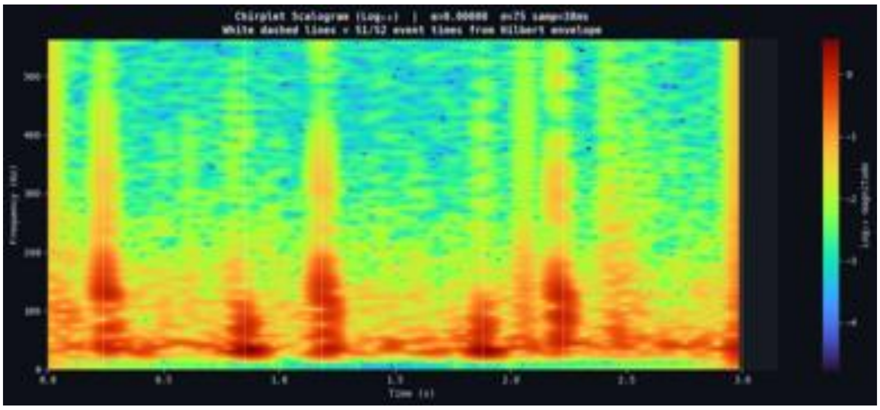


Fig.6 POST CHIRPLET TRANSFORMED PCG

Advanced extension of the Wavelet Transform designed to analyse signals whose frequency changes over time—a phenomenon known as chirping.

Heart murmurs and certain valvular pathologies exhibit "frequency modulations" where the pitch rises or falls within a single cardiac event.

Chirplet can **rotate and shear** in the time-frequency plane, making it more precise for tracking the rapidly changing dynamics of blood flow turbulence results in fig 6.

Equation used are as follows:

1. $T_{\alpha,\sigma}(k, n_0) = \sum_{n=1}^{N} z\,(n) e^{-j\frac{2\pi kn}{N}} \Psi^{*}_{\sigma,\alpha,n_0}(n)$→ z(n) is analytic signal from Hilbert transform.
2. $\Psi_{\sigma,\alpha,n_0}(n) = w_\sigma(n - n_0) e^{-j\frac{\alpha}{2}(n-n_0)^2}$

Features:

Validation Accuracy: 91.1%

Sensitivity: 98.5%

Specificity: 82.95

5.4.5 Constan Q Transform:

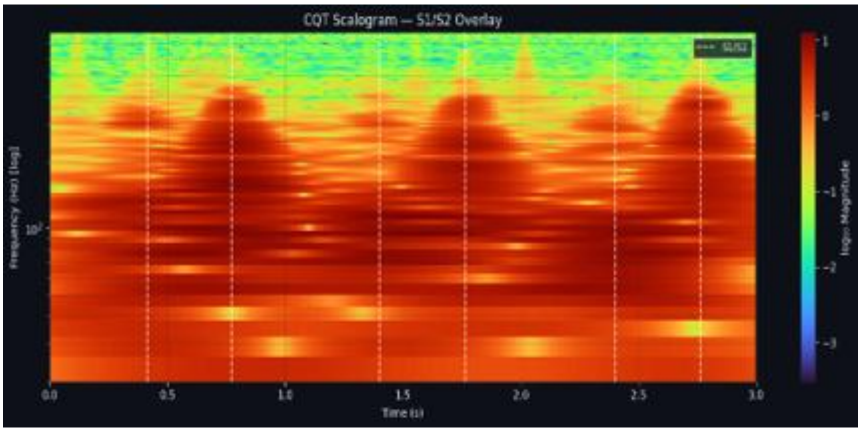


Fig.7 POST CQT TRANSFORMED PCG

Time-frequency representation where the frequency bins are geometrically spaced (logarithmic), and the ratio of centre frequency to bandwidth—known as the **Quality Factor (Q)**—remains constant.

It mimics the human ear's logarithmic perception of pitch.

It provides **high frequency resolution** for low-pitched sounds and **high temporal resolution** for high-frequency transients, results in **fig 7**.

Equation used are as follows:

1. $Q = \frac{f_k}{\Delta f_k} = \left(2^{1/B} - 1\right)^{-1}$ → Q is quality factor, f_k is centre frequency and $\triangle f_k$ is bandwidth.

2. $N_k = Q \cdot \frac{f_s}{f_k}$ → N_k is variable window length.

3. $X_{CQT}[k] = \frac{1}{N_k} \sum_{n=0}^{N_k-1} x\,[n] \cdot w[n,k] \cdot e^{-j2\pi nQ/N_k}$ →CQT equation, w[n,k] is hamming window.

Features:

Validation accuracy: 96.8%

Sensitivity: 95.7%

Specificity: 96.4%

5.5 SPIKE ENCODING:

3 types of encoding methods have been reviewed:

1. Latency encoding.
2. Delta encoding.
3. Rate encoding.

Figure 8 enlists a short difference description among these Encoding techniques.

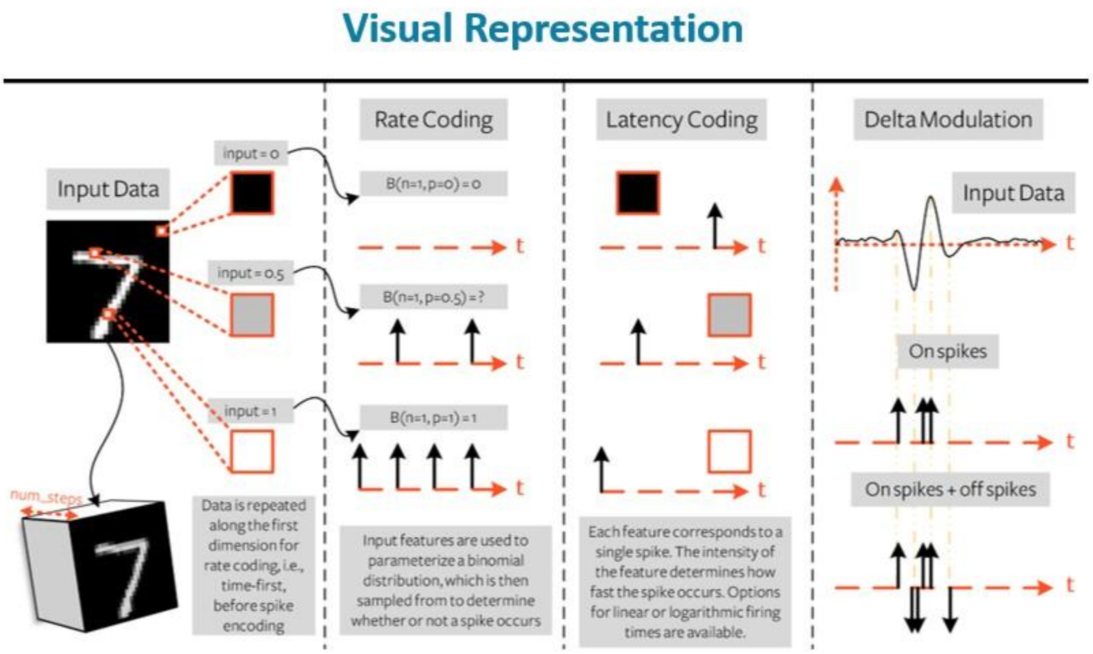


Fig.8 Latency, Delta, Rate encoding comparison

5.5.1 LATENCY ENCODING

Encodes information based on the timing or arrival of the first spike.

Maps the input intensity to a specific delay.

Strong i/p results in earlier spikes

$$t_{spike} = \tau \ln\left(\frac{x}{x-\theta}\right)$$

Linear approximation:

$$t_{spike} = T_{max} \cdot (1 - x_{norm})$$

5.5.2 DELTA ENCODING

Focuses on encoding the **changes** or differences in the input signal's intensity over time rather than its absolute value.

Generates spikes based on the change in the signal relative to a set threshold.

A spike S(t) is generated if:

$$|x(t) - x(t_{prev})| > \Delta_{threshold}$$

Positive Spike (+1) : if $x(t) - x(t_{prev}) > \Delta$

Negative Spike (-1) : if $x(t) - x(t_{prev}) < -\Delta$

5.5.3 RATE ENCODING

Represents information through the frequency of spikes within a specific time window.

Uses a Bernoulli trial mechanism where probability of a spike occurring at any discrete time step ‘t’ is proportional to the normalized intensity of the input feature ‘x[n]’.

$$P(S(t) = 1) = x_{norm}$$

Out of all these encoding techs, feature extractions coupled along with **rate encoding** provided the **best results**, hence rate encoding is being used as the spike encoding tech for the thesis. Provided below is the result of **validation accuracy** and **confusion matrices and sensitivity vs specificity** of different features extraction using rate encoding:

5.6 1D CSNN:

A specialized class of SNN designed to process one-dimensional temporal data like PCG signals.

Mimics biological neural systems by communicating through discrete, asynchronous events known as spikes.

As to why 1D CSNN is considered for this thesis, reasons are listed below

.1 1D kernels slide across the temporal axis to extract local features like sharp transients of S1/S2 heart sounds or frequency modulation of murmurs.

.2 The network maintains a significantly lower parameter count, hence suitable for edge-device deployment.

.3 Core processing unit of the network is the **LIF** neuron model, which consume significant power only during firing, the CSNN is highly efficient. Hence energy efficient.

.4 Aligns with neuromorphic hardware leading the way for low-power, wearable cardiac monitoring devices.

Detail as to why 1d CSNN is being used here instead of other existing SNN model i.e. highlighting its benefits and advantage are tabulated below

5.7 Evaluation metric:

Throughout the project to decide which encoding techniques or feature extraction is best aligned with our interest, we have used these 6-evaluation metric for this purpose:

1. **Validation Accuracy**: This is the primary metric used to compare the effectiveness of different feature extraction encoders.
2. **Sensitivity (Abnormal Detection)**: Measures the system's ability to correctly identify "Abnormal" heart sound recordings.
3. **Specificity**: Measures the system's ability to correctly identify "Normal" heart sound recordings.
4. **AUC-ROC Score**: Used to evaluate the model's ability to distinguish between classes across different threshold settings.
5. **Train-Val Gap**: Employed as a measure of generalization to identify risks of **overfitting** or **underfitting** for each encoder.
6. **Confusion Matrix**: Utilized to provide a detailed breakdown of classification results across all five encoder methods.

Provided below is the table 1 of different feature extraction based on the above-mentioned evaluation metric:

ENCODER	ACCURACY	AUC-ROC	SENSITIVITY	SPECIFICITY	TRAIN VAL-GAP
CQT	96.80%	96.10%	95.70%	96.40%	+2.2%
MEL	94.50%	93.50%	96.10%	90.70%	-0.5%
CHIRPLET	91.10%	90.80%	98.60%	82.90%	+11.0%
CWT	87.30%	85.70%	88.60%	82.50%	+6.7%
MFCC	73.60%	69.10%	73.90%	63.90%	-0.4%

Table. 1 Trade-off between Extraction Encoder against EVAL Metric

CHAPTER 6
HARDWARE IMPLEMENTATION

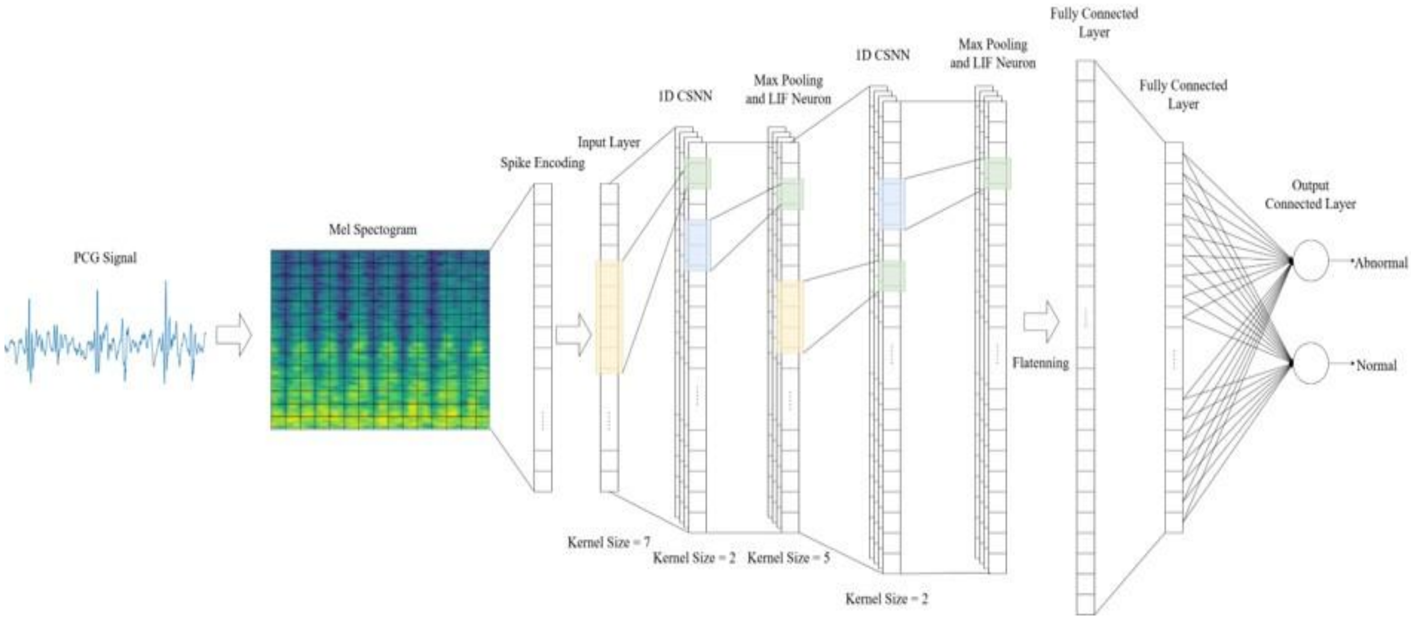


Fig. 9 1D conv block diagram

Two 1D conv blocks (24→32→64) with Batch Norm, LIF neurons, and Max pooling → Fully connected layer (2 classes) as shown in **Figure 9.**
Total Parameters 16K, LIF (β=0.9) with fast sigmoid gradient (slope=25)

The hardware implementation phase of this project focuses on transitioning the 1D Convolutional Spiking Neural Network (CSNN) from software simulation to a low power, real time diagnostic hardware platform (targeting FPGA/ASIC). The primary goal is to leverage the **asynchronous** and **event driven** nature of spiking neurons to achieve high diagnostic accuracy with minimal energy consumption. FPGAs are selected for their ability to handle the parallel processing requirements of convolutional layers while maintaining a flexible RTL (Verilog) design for custom neuron dynamics. Xilinx Vivado Design Suite is used for this purpose.

The core of the hardware implementation is the **Leaky Integrate-and-Fire (LIF)** neuron module. In RTL, the LIF dynamics are implemented using:

1. **Membrane Potential Integration:** A digital accumulator that sums incoming weighted spikes.
2. **Leaky Mechanism:** Implemented as a bit-shift right operation, providing a computationally "cheap" division to simulate the decay of the membrane potential toward the resting state.

3. **Thresholding & Reset:** A comparator triggers an output spike (1-bit signal) when the potential exceeds a predefined threshold, followed by a hardware reset of the accumulator.

FLOW CHART – 3 RTL TO GDS

1D kernels slide across the temporal signal, significantly reducing the memory required to store synaptic weights (~120K parameters).

The 1-bit spiking activation replaces floating-point multiplications required in CNNs with add-and-accumulate operations, reducing the consumption of FPGA DSP slices.

The preprocessing pipeline (filtering and normalization) and the **CQT/Mel encoders** are implemented as a front-end digital signal processing (DSP) block.

Significant advantage of this hardware implementation is its event-driven power profile:

1. In a heart sound cycle, there are periods of silence between heartbeats.1D CSNN remains computationally inactive (zero spikes) during this interval.
2. Lean architecture ensures that the entire classification system can reside on a small-footprint chip, making it a viable solution for continuous, battery-operated cardiac monitoring.

The hardware performance will be benchmarked against these evaluation metrics:

1. **Inference Latency:** The time taken from raw PCG input to "Normal/Abnormal" classification.
2. **Dynamic Power:** Measured in milliwatts (mW) to demonstrate the neuromorphic advantage over traditional CPU/GPU-based inference.

Figure 10 provided below is the tape out for chip implementation and fabrication.

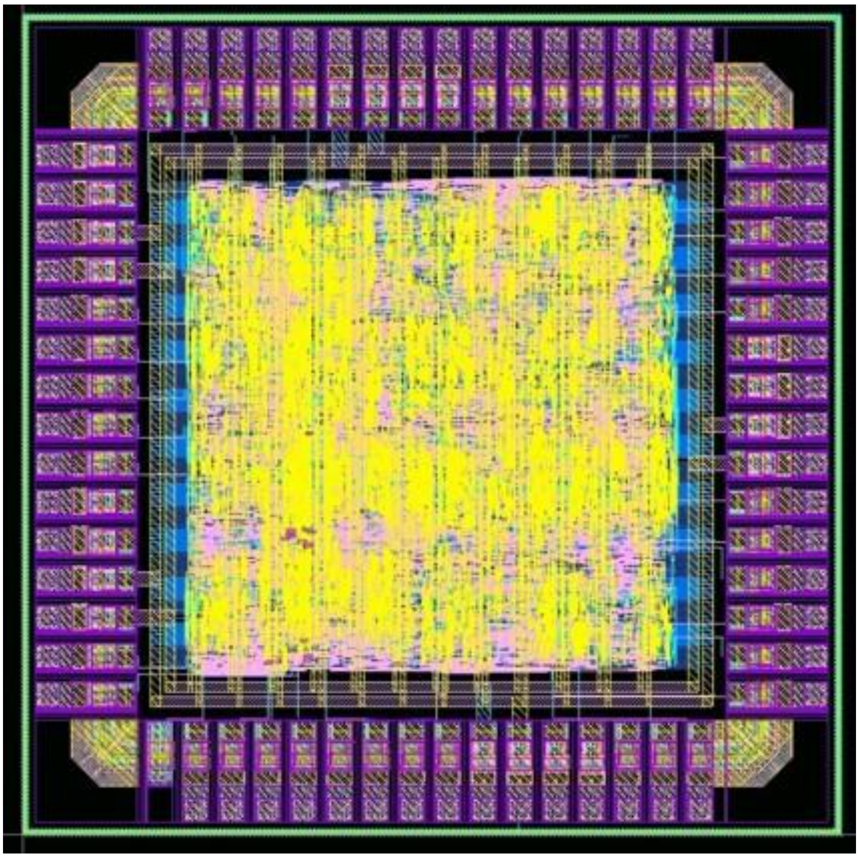


Fig. 10 tape out of chip

CHAPTER 7
RESULTS & DISCUSSION

7.1 TRAINING LOSS PLOT

Training loss measures the error between the model's predictions and the actual labels during the learning phase.

Decreasing loss curve → model successfully learning data pattern. **Figure 11** shows the training loss plot of each encoder.

The Chirplet transform showed the fastest convergence, reaching a remarkably low loss of ~0.02 at Epoch 100, though this carries a high risk of overfitting.

The CQT and Mel encoders provided a more balanced fit with losses of ~0.12 and ~0.22, respectively.

Higher final losses, such as MFCC at ~0.59, indicate a poorer fit for the specific spiking neural network architecture used.

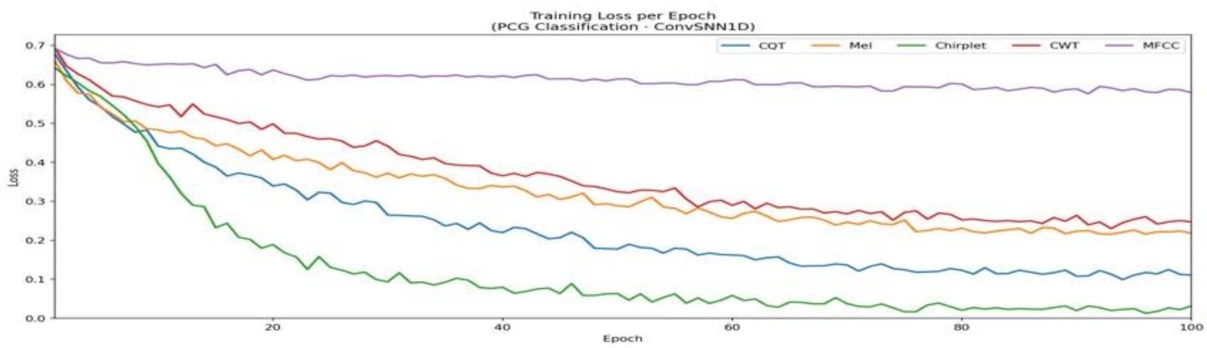


Fig. 11 Training Loss per epoch

7.2 SPECIFICITY VS SENSITIVITY

These two metrics define the diagnostic reliability of a medical classifier:

Sensitivity: Measures the ability of the model to correctly identify patients with a disease (True Positive rate). **Specificity**: Measures the ability of the model to correctly identify those without the disease (True Negative rate). **Figure 12** shows the sensitivity vs specificity trade off of each encoder.

The Chirplet transform achieved a peak sensitivity of 98.6%, while the CQT encoder maintained a robust 95.7%. While the Chirplet transform had high sensitivity, its lower specificity of 82.9% made it less ideal for screening than the CQT, which maintained a high specificity of 96.4%.

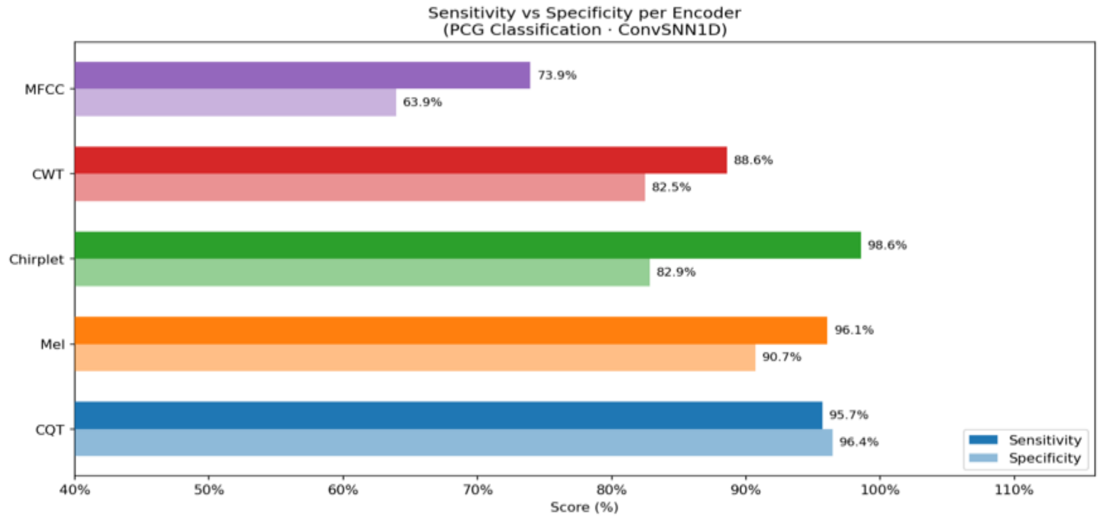


Fig. 12 Specificity vs Sensitivity per Encoder

7.3 VALIDATION ACCURACY

Validation accuracy is a metric that measures how well a neural network generalizes to unseen data. Scores indicate a model's capability to accurately classify heart valve disorders outside of its training set. In this project, it was used to evaluate the performance of five different feature extraction "encoders" when paired with the **1D CSNN** architecture. **Figure 13** provides the validation accuracy per encoder plot

The results showed that the **Constant-Q Transform (CQT)** achieved the highest validation accuracy at **96.8%**, while **Mel-spectrograms** followed at **94.5%**, significantly outperforming the **85%** accuracy threshold where previous frameworks failed in noisy environments.

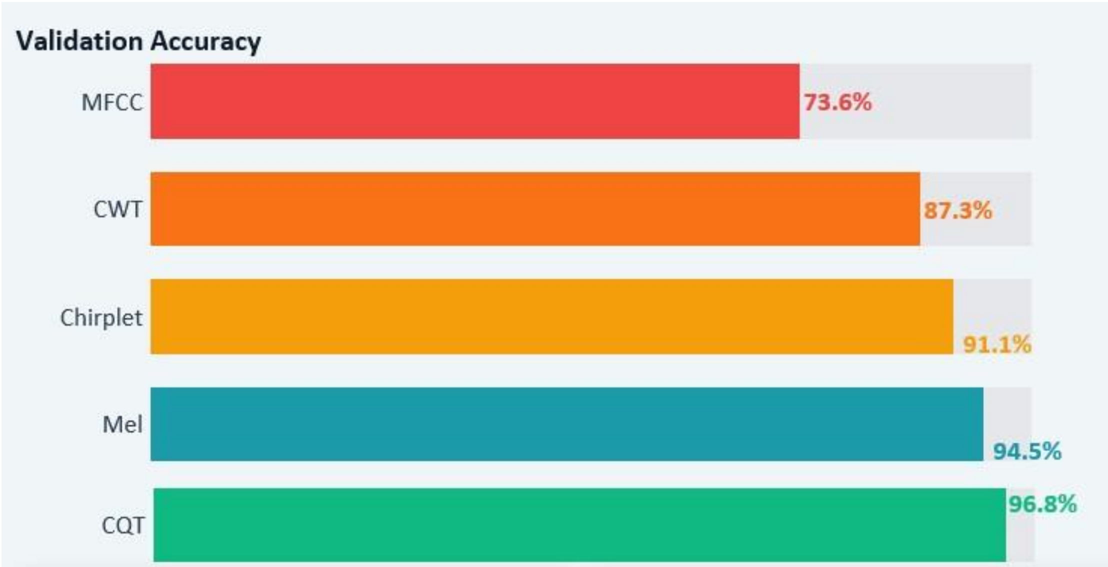


Fig. 13 Validation Accuracy per Encoder

CHAPTER 8
SUMMARY AND CONCLUSION

8.1 SUMMARY

Our project overcame these limitations by integrating robust signal encoding with a 1D CSNN architecture, achieving superior performance metrics across multiple encoders. As demonstrated in the results section, the Constant Q Transform (CQT) achieved the highest validation accuracy of 96.8%, followed closely by Mel spectrograms at 94.5%. This significantly outperforms the 85% accuracy threshold where previous PCT based models failed in noisy environments. The Chirplet transform also showed strong performance at 91.1%, while the Continuous Wavelet Transform (CWT) and MFCC trailed at 87.3% and 73.6%, respectively.

The stability of our framework is further evidenced by the training loss analysis. While the Chirplet transform exhibited the fastest convergence, reaching a remarkably low loss of ~0.02 at Epoch 100, it carries a high risk of overfitting. In contrast, the CQT and Mel encoders provide a more balanced fit with losses of ~0.12 and ~0.22, ensuring better generalization for real world deployment. The CWT and MFCC demonstrated higher final losses of ~0.25 and ~0.59, indicating a poorer fit for this specific SNN architecture.

The research systematically evaluated five distinct feature extraction encoders to determine the most effective input representation for the spiking architecture. The preprocessing pipeline, comprising DC removal, notch filtering, and bandpass filtering (25–400 Hz), ensured the signal was optimized for the Leaky Integrate-and-Fire (LIF) neurons. The final architecture remains remarkably lean with only ~120,000 parameters, maintaining a low computational footprint without compromising performance.

By successfully bridging the gap between high fidelity acoustic analysis and low power hardware constraints, this project delivers a state-of-the-art diagnostic tool optimized for the next generation of medical wearables. We have demonstrated that a 1D CSNN architecture, when paired with CQT encoding, provides a clinically reliable balance of high sensitivity (95.7%) and specificity (96.4%) that surpasses the limitations of traditional, power-intensive models. This framework not only achieves superior accuracy on noisy, real-world datasets but also operates with a minimal parameter count, making it a viable solution for real-time, automated heart valve disease classification on edge devices.

8.2 KEY FINDINGS AND CONCULSION

Superiority of CQT: The Constant-Q Transform (CQT) was the most effective encoder, achieving a peak Validation Accuracy of 96.8% and an AUC-ROC of 96.1%. Its logarithmic frequency resolution naturally aligns with the harmonic nature of heart sounds. Robustness of Mel Encoding: While CQT provided the highest accuracy, the Mel Spectrogram demonstrated the best generalization with a minimal Train-Val Gap of -0.5%, i.e. highly resistant to overfitting.

Sensitivity vs. Specificity: The Chirplet Transform yielded the highest Sensitivity (98.6%), making it a valuable for screening applications where missing an Abnormal case is the primary risk. Hardware Feasibility: CSNN allows for significant power savings compared to CNN. The implementation of 1-bit spiking activation and fixed-point arithmetic makes the model ideal for FPGA deployment via Vivado, where it can operate with high energy efficiency during cardiac silences.

8.3 FUTURE WORK

Hardware-in-the-Loop (HIL) Testing: Moving beyond simulation to measure actual power consumption (mW) and latency on physical FPGA hardware (e.g., Xilinx Artix-7).
Multiclass Classification: Expanding the model from binary (Normal/Abnormal) to specific valvular disease identification (e.g., Mitral Regurgitation, Aortic Stenosis).
Real-time Wearable Integration: Optimizing the RTL design for a "Tape-out" ready chip that can be integrated into a smart stethoscope or a continuous monitoring patch.
FPGA implementation where we measure real energy consumption vs GPU baseline.
Cross dataset validation: Test on PhysioNet 2022, CirCor datasets for generalization of study.

CHAPTER 9

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